

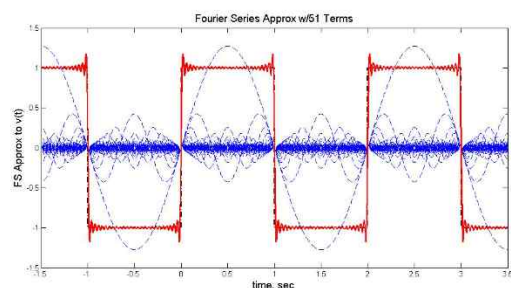
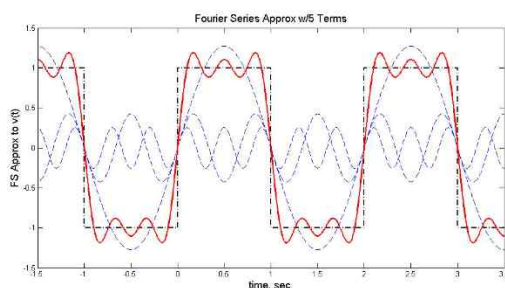
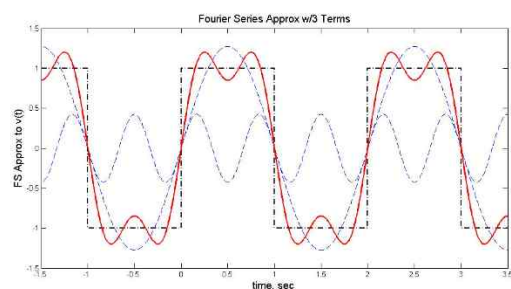
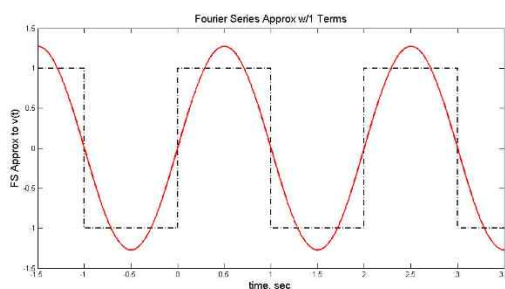
Chap 3. Fourier Representations for Periodic Signals

- Continuous-Time Periodic Signals
- Continuous-Time Fourier Series (CTFS)
- Discrete-Time Periodic Signals
- Discrete-Time Fourier Series (DTFS)
- Gibbs Phenomenon
- Properties of Fourier Representation

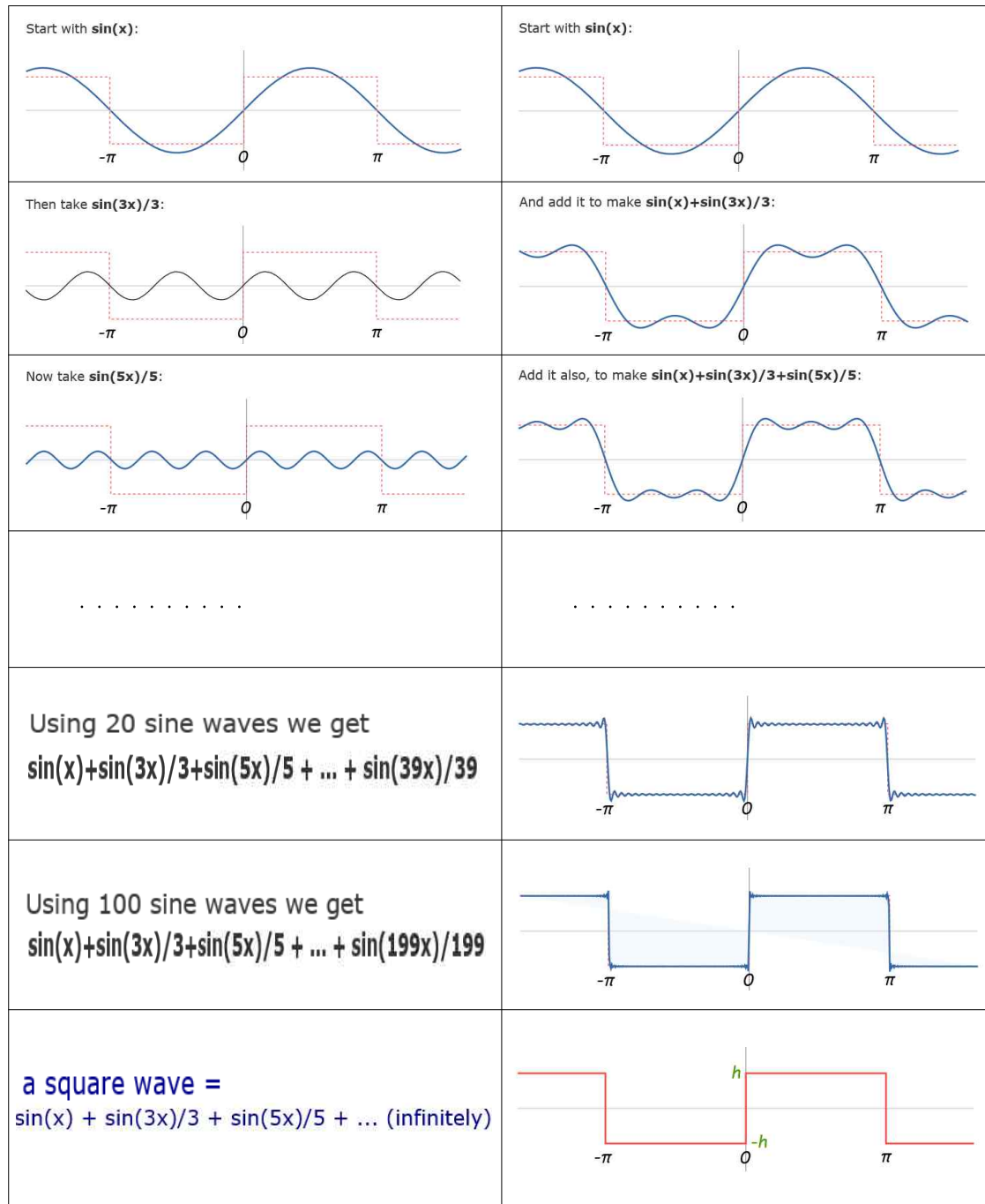
- Discrete-Time Nonperiodic Signals
- Discrete-Time Fourier Transform (DTFT)
- Continuous-Time Nonperiodic Signals
- Continuous-Time Fourier Transform (CTFT)

- Discrete Fourier Transform (DFT)
- Fast Fourier Transform (FFT)

(Example)



※ Can we make a square wave by adding sinusoidal waves? [YES !!]



§3.1 A Historical Perspective

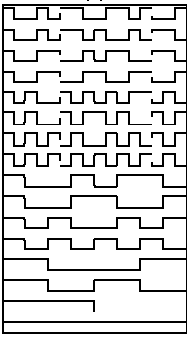
- Babylonians
 - L. Euler (1748)
 - D. Bernoulli (1753)
 - J. L. Lagrange (1759)
 - Jean Baptiste Joseph Fourier (1768-1830)
Fourier transform / heat propagation and diffusion
 - Pierre Simon de Laplace (1749-1827)
Laplace transform / astronomy
- ※ Read the section 1 of the chapter 3

§3.2 The response of LTI systems to complex exponentials

- an advantage in the study of LTI systems :
A signal can be represented as linear combinations
of basic signals.

$$(\text{a signal}) = \sum (\text{constant}) \times (\text{basic signal}) \quad (\Delta)$$

- some examples of basic signals

series	basic components (signals)	remarks
Taylor series	$x^0, x^1, x^2, x^3, \dots$	$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$
Walsh function		
3-D vector	$(\hat{x}, \hat{y}, \hat{z}), (\hat{r}, \hat{\phi}, \hat{z}), (\hat{r}, \hat{\phi}, \hat{\theta})$	
complex numbers	$(1, j)$	
colors	(red, yellow, blue)	green = 0 r + 0.5 y + 0.5 b
sifting	$\delta[n], \delta[n \pm 1], \delta[n \pm 2], \dots$	$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$

- responses of LTI systems to complex exponentials

	continuous-time	discrete-time
complex exponential input	$x(t) = e^{st}$	$x[n] = z^n$
output to the complex exponential input	$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$ $= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau$ $= e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$ $\equiv H(s) e^{st}$	$y[n] = \sum_{k=-\infty}^{+\infty} h[k] x[n-k]$ $= \sum_{k=-\infty}^{+\infty} h[k] z^{n-k}$ $= z^n \sum_{k=-\infty}^{+\infty} h[k] z^{-k}$ $\equiv H(z) z^n$
transfer function	$H(s) = \int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau$	$H(z) = \sum_{k=-\infty}^{+\infty} h[k] z^{-k}$
	The output of the system is a complex exponential as the input multiplied by the complex constant number $H(s)$, which is not a function of time, t , but is only a function of frequency, s .	The output of the system is a complex exponential as the input multiplied by the complex constant number $H(z)$, which is not a function of time, n , but is only a function of frequency, z .
the complex sinusoidal input	$x(t) = e^{j\omega t}$	$x[n] = e^{j\omega n}$
frequency response	$H(j\omega) = \int_{-\infty}^{+\infty} h(\tau) e^{-j\omega\tau} d\tau$	$H(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} h[k] e^{-jk\omega}$
output to the unit amplitude complex sinusoidal input	$y(t) = e^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$	$y[n] = e^{j\omega n} \sum_{k=-\infty}^{+\infty} h[k] e^{-jk\omega}$
	The output of the system is a complex sinusoid of the <u>same frequency</u> as the input multiplied by the complex constant number $H(j\omega)$, that is not a function of time, t , but is only a function of frequency, ω .	The output of the system is a complex sinusoid of the <u>same frequency</u> as the input multiplied by the complex constant number $H(e^{j\omega})$, that is not a function of time, n , but is only a function of frequency, ω .

§ EM 7.8 (p.401) Transfer Function / EM 10.3 Fourier Series
 C&L 13.4 (p.482) 13.6 (p.487) / C&L 14.2 (p.526)

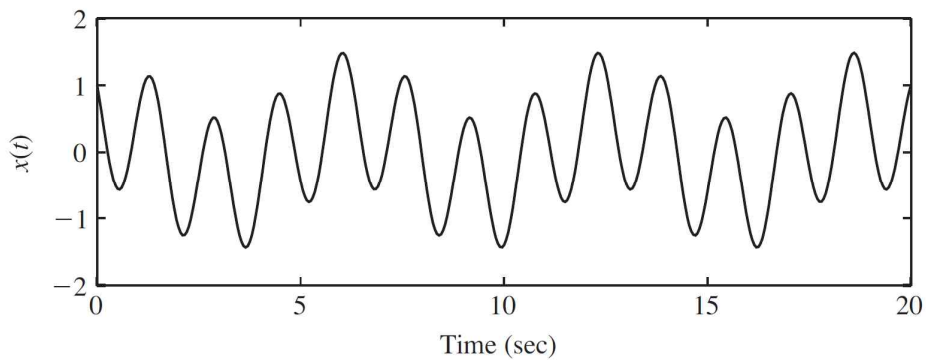
- the input-output of a LTI system
(input: summation of several sinusoidal waves,
i.e., a linear combination of sinusoidal waves)

$x(t) \rightarrow \boxed{h(t)} \rightarrow y(t)$	
$e^{j\omega_1 t} \rightarrow H(j\omega_1) e^{j\omega_1 t}$ $a_1 e^{j\omega_1 t} \rightarrow a_1 H(j\omega_1) e^{j\omega_1 t}$ $e^{j\omega_2 t} \rightarrow H(j\omega_2) e^{j\omega_2 t}$ $a_2 e^{j\omega_2 t} \rightarrow a_2 H(j\omega_2) e^{j\omega_2 t}$ $a_1 e^{j\omega_1 t} + a_2 e^{j\omega_2 t} \rightarrow a_1 H(j\omega_1) e^{j\omega_1 t} + a_2 H(j\omega_2) e^{j\omega_2 t}$ $\sum_{k=1}^M a_k e^{j\omega_k t} \rightarrow \sum_{k=1}^M a_k H(j\omega_k) e^{j\omega_k t}$	
$e^{s_1 t} \rightarrow H(s_1) e^{s_1 t}$ $a_1 e^{s_1 t} \rightarrow a_1 H(s_1) e^{s_1 t}$ $e^{s_2 t} \rightarrow H(s_2) e^{s_2 t}$ $a_2 e^{s_2 t} \rightarrow a_2 H(s_2) e^{s_2 t}$ $a_1 e^{s_1 t} + a_2 e^{s_2 t} \rightarrow a_1 H(s_1) e^{s_1 t} + a_2 H(s_2) e^{s_2 t}$ $\sum_{k=1}^M a_k e^{s_k t} \rightarrow \sum_{k=1}^M a_k H(s_k) e^{s_k t}$	

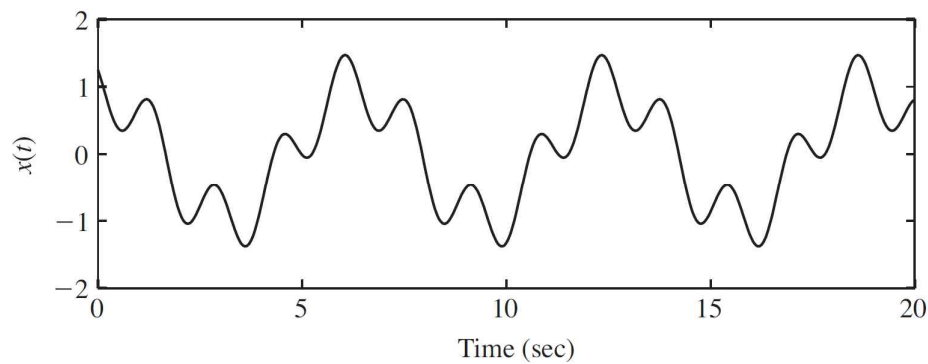
● Sum of Sinusoids

$$x(t) = A_1 \cos(t) + A_2 \cos(4t + \pi/3) + A_3 \cos(8t + \pi/2)$$

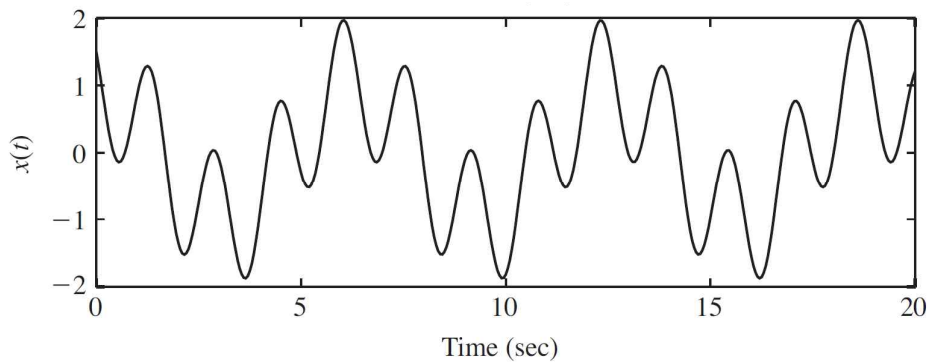
(a) $x(t) = 0.5 \cos(t) + 1 \cos(4t + \pi/3) + 0 \cos(8t + \pi/2)$



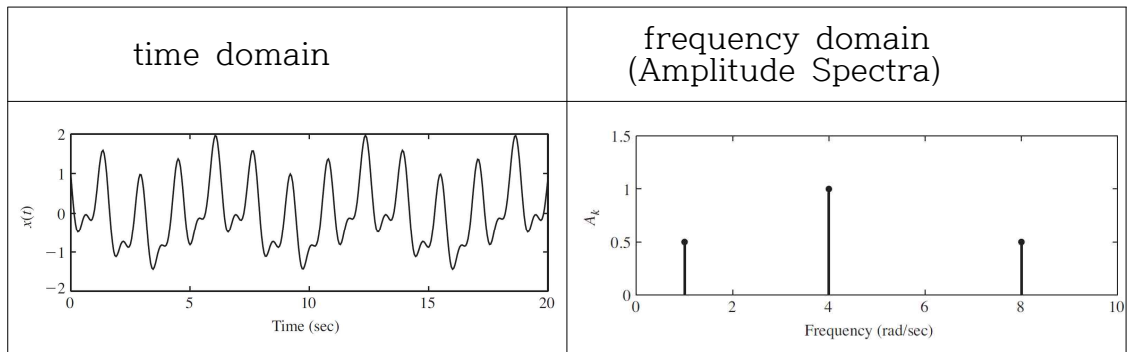
(b) $x(t) = 1 \cos(t) + 0.5 \cos(4t + \pi/3) + 0 \cos(8t + \pi/2)$



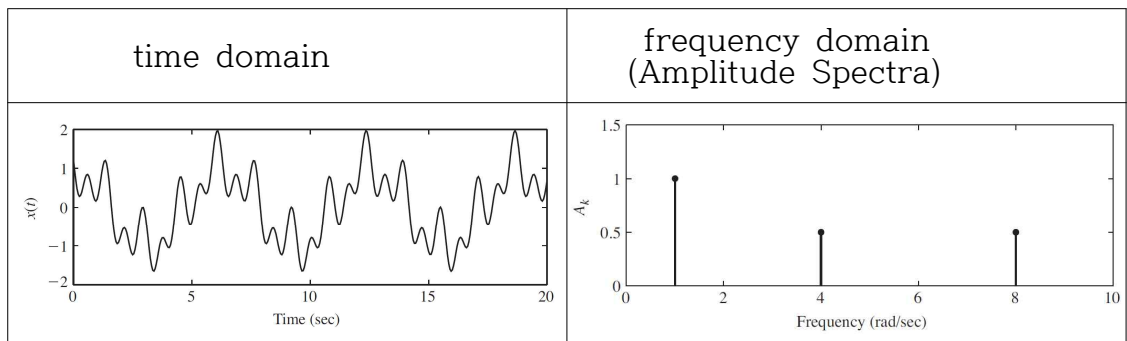
(c) $x(t) = 1 \cos(t) + 1 \cos(4t + \pi/3) + 0 \cos(8t + \pi/2)$



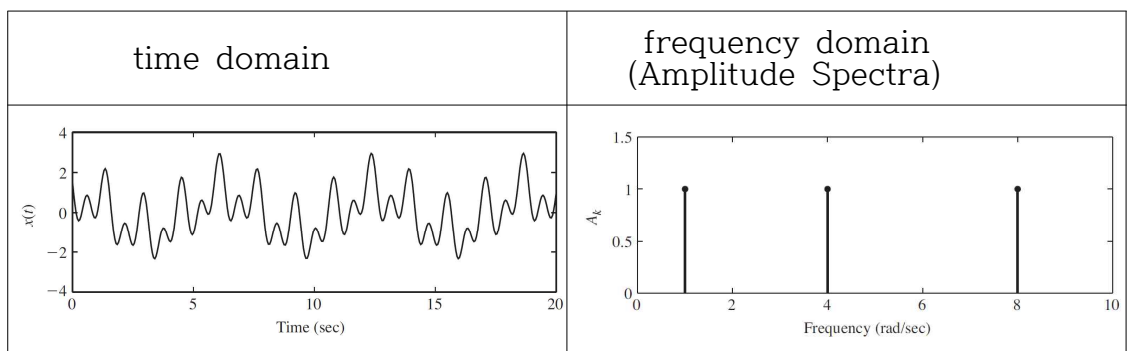
(d) $x(t) = 0.5 \cos(t) + 1 \cos(4t + \pi/3) + 0.5 \cos(8t + \pi/2)$



(e) $x(t) = 1 \cos(t) + 0.5 \cos(4t + \pi/3) + 0.5 \cos(8t + \pi/2)$



(f) $x(t) = 1 \cos(t) + 1 \cos(4t + \pi/3) + 1 \cos(8t + \pi/2)$



(Example) synthetic vowel

$$x(t) = \sum_{k=-N}^N a_k e^{j2\pi k f_0 t}$$

- synthesis of the vowel, /ah/

Table 3-1: Complex amplitudes for the periodic signal that approximates the vowel “ah”. The a_k coefficients are given for positive indices k , but the values for negative k are the conjugates, $a_{-k} = a_k^*$.

k	f_k (Hz)	a_k	Mag	Phase
1	100	0	0	0
2	200	$386 + j6101$	6113	1.508
3	300	0	0	0
4	400	$-4433 + j14024$	14708	1.877
5	500	$24000 - j4498$	24418	-0.185
6	600	0	0	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
15	1500	0	0	0
16	1600	$828 - j6760$	6811	-1.449
17	1700	$2362 + j0$	2362	0

- spectrum of /ah/

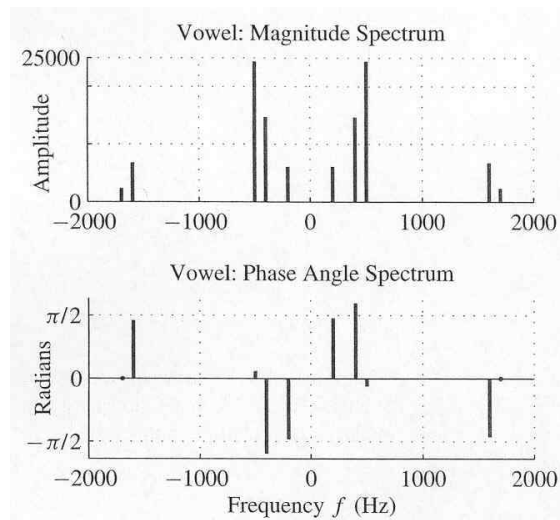
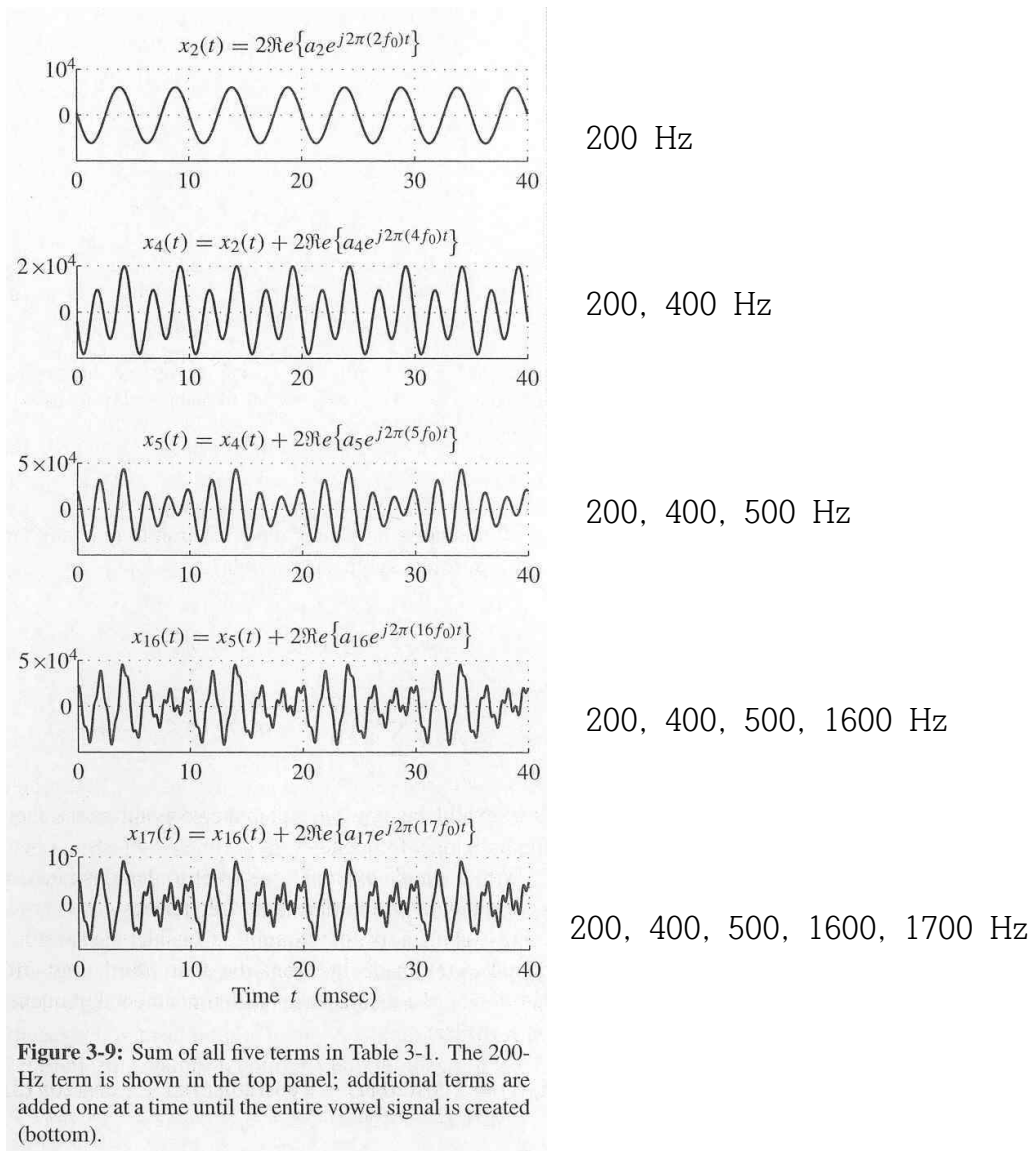


Figure 3-8: Spectrum of signal defined in Table 3-1. The magnitude is an even function with respect to $f = 0$; the phase is odd.

- synthesizing of /ah/ [waveforms in time domain]



- Real Waveform of /ah/



§3.3 Fourier series representation of continuous-time periodic signals

※ Circuit and Lab Textbook

Harmonics

Some feeling for the validity of representing a general *periodic* function by an infinite sum of sine and cosine functions may be gained by considering a simple example. Let us first assume a cosine function of radian frequency ω_0 ,

$$v_1(t) = 2 \cos \omega_0 t$$

where

$$\omega_0 = 2\pi f_0$$

and the period T is

$$T = \frac{1}{f_0} = \frac{2\pi}{\omega_0}$$

Although T does not usually carry a zero subscript, it is the period of the fundamental frequency. The **harmonics** of this sinusoid have frequencies $n\omega_0$, where ω_0 is the fundamental frequency and $n = 1, 2, 3, \dots$. The frequency of the first harmonic is the **fundamental frequency**.

Next let us combine this with a third-harmonic voltage given by

$$v_3(t) = \cos 3\omega_0 t$$

The fundamental $v_1(t)$, the third harmonic $v_3(t)$, and the sum of these two waves $v(t)$ are shown as functions of time in Fig. 17.1a. Note that the sum is also periodic, with period $T = 2\pi/\omega_0$.

The form of the resultant periodic function changes as the phase and amplitude of the third-harmonic component change. Thus, Fig. 17.1b shows the effect of combining $v_1(t)$ and a third harmonic of slightly larger amplitude,

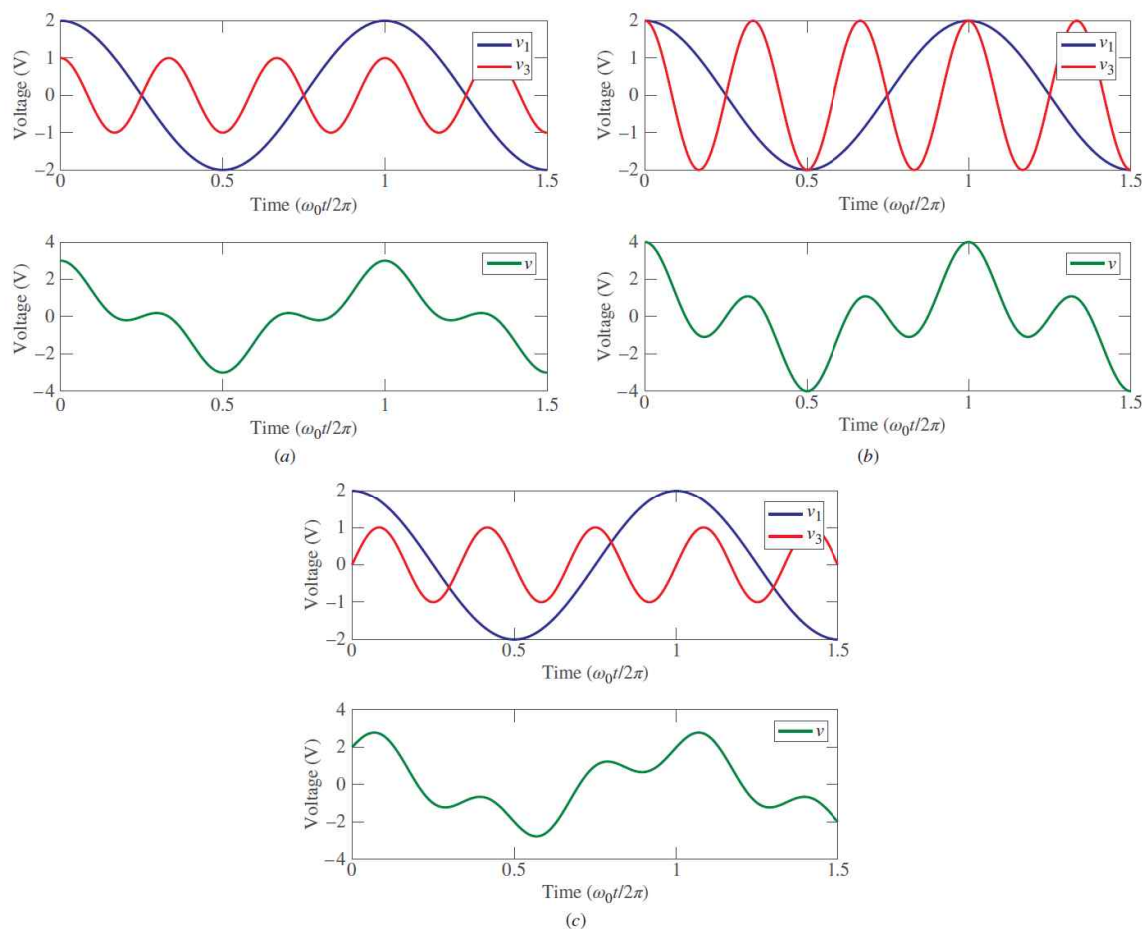
$$v_3(t) = 2 \cos 3\omega_0 t$$

By shifting the phase of the third harmonic by 90 degrees to give

$$v_3(t) = \sin 3\omega_0 t$$

the sum, shown in Fig. 17.1c, takes on a still different character. In all cases, the period of the resultant waveform is the same as the period of the fundamental waveform. The nature of the waveform depends on the amplitude and phase of every possible harmonic component, and we will find that we can generate waveforms which have extremely nonsinusoidal characteristics by an appropriate combination of sinusoidal functions.

After we have become familiar with the use of the sum of an infinite number of sine and cosine functions to represent a periodic waveform, we will consider the frequency-domain representation of a general nonperiodic waveform in a manner similar to the Laplace transform.



■ **FIGURE 17.1** Several example waveforms combining a fundamental and a third harmonic. The fundamental is $v_1(t) = 2 \cos \omega_0 t$, and the third harmonic is $v_3(t) = \cos 3\omega_0 t$; (b) $v_3(t) = 2 \cos 3\omega_0 t$; (c) $v_3(t) = \sin 3\omega_0 t$.

※ Some Useful Trigonometric Identities

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\cos(\alpha \pm 90^\circ) = \mp \sin \alpha$$

$$\sin(\alpha \pm 90^\circ) = \pm \cos \alpha$$

$$\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta)$$

$$\sin \alpha \sin \beta = \frac{1}{2} \cos(\alpha - \beta) - \frac{1}{2} \cos(\alpha + \beta)$$

$$\sin \alpha \cos \beta = \frac{1}{2} \sin(\alpha + \beta) + \frac{1}{2} \sin(\alpha - \beta)$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\sin^2 \alpha = \frac{1}{2}(1 - \cos 2\alpha)$$

$$\cos^2 \alpha = \frac{1}{2}(1 + \cos 2\alpha)$$

$$\sin \alpha = \frac{e^{j\alpha} - e^{-j\alpha}}{j2}$$

$$\cos \alpha = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$$

$$e^{\pm j\alpha} = \cos \alpha \pm j \sin \alpha$$

$$A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \cos \left(\alpha + \tan^{-1} \frac{-B}{A} \right)$$

● The Fourier Series

Given such a periodic function $f(t)$, the Fourier theorem states that $f(t)$ may be represented by the infinite series

$$\begin{aligned} f(t) &= a_0 + a_1 \cos \omega_0 t + a_2 \cos 2\omega_0 t + \cdots \\ &\quad + b_1 \sin \omega_0 t + b_2 \sin 2\omega_0 t + \cdots \\ &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \end{aligned} \quad [1]$$

where the fundamental frequency ω_0 is related to the period T by

$$\omega_0 = \frac{2\pi}{T}$$

and where a_0 , a_n , and b_n are constants that depend upon n and $f(t)$. Equation [1] is the trigonometric form of the **Fourier series** for $f(t)$, and the process of determining the values of the constants a_0 , a_n , and b_n is called *Fourier analysis*. Our object is not the proof of this theorem, but only a simple development of the procedures of Fourier analysis and a feeling that the theorem is plausible.

Some Useful Trigonometric Integrals

Before we discuss the evaluation of the constants appearing in the Fourier series, let us collect a set of useful trigonometric integrals. We let both n and k represent any element of the set of integers 1, 2, 3, In the following integrals, 0 and T are used as the integration limits, but it is understood that any interval of one period is equally correct.

$$\int_0^T \sin n\omega_0 t \, dt = 0 \quad [2]$$

$$\int_0^T \cos n\omega_0 t \, dt = 0 \quad [3]$$

$$\int_0^T \sin k\omega_0 t \cos n\omega_0 t \, dt = 0 \quad [4]$$

$$\int_0^T \sin k\omega_0 t \sin n\omega_0 t \, dt = 0 \quad (k \neq n) \quad [5]$$

$$\int_0^T \cos k\omega_0 t \cos n\omega_0 t \, dt = 0 \quad (k \neq n) \quad [6]$$

Those cases which are excepted in Eqs. [5] and [6] are also easily evaluated; we obtain

$$\int_0^T \sin^2 n\omega_0 t \, dt = \frac{T}{2} \quad [7]$$

$$\int_0^T \cos^2 n\omega_0 t \, dt = \frac{T}{2} \quad [8]$$

Evaluation of the Fourier Coefficients

The evaluation of the unknown constants in the Fourier series may now be accomplished readily. We first attack a_0 . If we integrate each side of Eq. [1] over a full period, we obtain

$$\int_0^T f(t) \, dt = \int_0^T a_0 \, dt + \int_0^T \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \, dt$$

But every term in the summation is of the form of Eq. [2] or [3], and thus

$$\int_0^T f(t) \, dt = a_0 T$$

or

$$a_0 = \frac{1}{T} \int_0^T f(t) \, dt \quad [9]$$

This constant a_0 is simply the average value of $f(t)$ over a period, and we therefore describe it as the dc component of $f(t)$.

To evaluate one of the cosine coefficients, for example, a_k , the coefficient of $\cos k\omega_0 t$, we first multiply each side of Eq. [1] by $\cos k\omega_0 t$ and then integrate both sides of the equation over a full period:

$$\begin{aligned} \int_0^T f(t) \cos k\omega_0 t \, dt &= \int_0^T a_0 \cos k\omega_0 t \, dt \\ &+ \int_0^T \sum_{n=1}^{\infty} a_n \cos k\omega_0 t \cos n\omega_0 t \, dt \\ &+ \int_0^T \sum_{n=1}^{\infty} b_n \cos k\omega_0 t \sin n\omega_0 t \, dt \end{aligned}$$

From Eqs. [3], [4], and [6] we note that every term on the right-hand side of this equation is zero except for the single a_n term where $k = n$. We evaluate that term using Eq. [8], and in so doing we find a_k , or a_n :

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega_0 t \, dt \quad [10]$$

This result is *twice* the average value of the product $f(t) \cos n\omega_0 t$ over a period.

In a similar way, we obtain b_k by multiplying by $\sin k\omega_0 t$, integrating over a period, noting that all but one of the terms on the right-hand side are zero, and performing that single integration by Eq. [7]. The result is

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega_0 t \, dt \quad [11]$$

which is *twice* the average value of $f(t) \sin n\omega_0 t$ over a period.

Equations [9] to [11] now enable us to determine values for a_0 and all the a_n and b_n in the Fourier series, Eq. [1], as summarized below:

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \quad [1]$$

$$\omega_0 = \frac{2\pi}{T} = 2\pi f_0$$

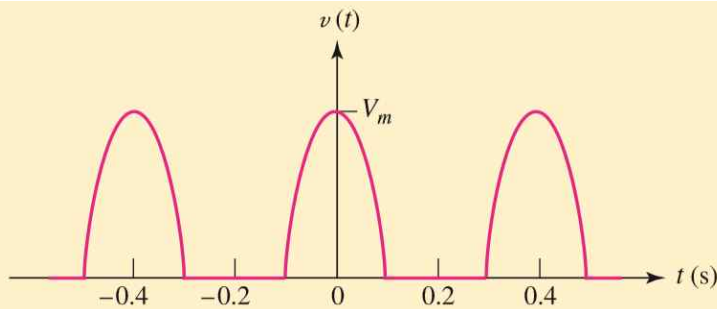
$$a_0 = \frac{1}{T} \int_0^T f(t) \, dt \quad [9]$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega_0 t \, dt \quad [10]$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega_0 t \, dt \quad [11]$$

(Example)

The “half-sinusoidal” waveform shown in Fig. 18.2 represents the voltage response obtained at the output of a half-wave rectifier circuit, a nonlinear circuit whose purpose is to convert a sinusoidal input voltage to a (pulsating) approximation to dc. Find the Fourier series representation of this waveform.



► **Collect the known information.**

In order to represent this voltage as a Fourier series, we must first determine the period and then express the graphical voltage as an analytical function of time. From the graph, the period is seen to be

$$T = 0.4 \text{ s}$$

and thus

$$f_0 = 2.5 \text{ Hz}$$

and

$$\omega_0 = 5\pi \text{ rad/s}$$

► **Devise a plan.**

The most straightforward approach is to apply Eqs. [9] to [11] to calculate the set of coefficients a_0 , a_n , and b_n . To do this, we need a functional expression for $v(t)$, the most straightforward being defined over the interval $t = 0$ to $t = 0.4$ as

$$v(t) = \begin{cases} V_m \cos 5\pi t & 0 \leq t \leq 0.1 \\ 0 & 0.1 \leq t \leq 0.3 \\ V_m \cos 5\pi t & 0.3 \leq t \leq 0.4 \end{cases}$$

However, choosing the period to extend from $t = -0.1$ to $t = 0.3$ will result in fewer equations and, hence, fewer integrals:

$$v(t) = \begin{cases} V_m \cos 5\pi t & -0.1 \leq t \leq 0.1 \\ 0 & 0.1 \leq t \leq 0.3 \end{cases} \quad [12]$$

This form is preferable, although either description will yield the correct results.

► **Construct an appropriate set of equations.**

The zero-frequency component is easily obtained:

$$a_0 = \frac{1}{0.4} \int_{-0.1}^{0.3} v(t) dt = \frac{1}{0.4} \left[\int_{-0.1}^{0.1} V_m \cos 5\pi t dt + \int_{0.1}^{0.3} 0 dt \right]$$

The amplitude of a *general* cosine term is

$$a_n = \frac{2}{0.4} \int_{-0.1}^{0.1} V_m \cos 5\pi t \cos 5\pi n t dt$$

and the amplitude of a general sine term is

$$b_n = \frac{2}{0.4} \int_{-0.1}^{0.1} V_m \cos 5\pi t \sin 5\pi n t dt$$

which, in fact, is always zero, and hence will not be considered further.

► **Determine if additional information is required.**

The form of the function we obtain upon integrating is different when n is unity than it is for any other choice of n . If $n = 1$, we have

$$a_1 = 5V_m \int_{-0.1}^{0.1} \cos^2 5\pi t dt = \frac{V_m}{2} \quad [13]$$

whereas if n is not equal to unity, we find

$$a_n = 5V_m \int_{-0.1}^{0.1} \cos 5\pi t \cos 5\pi n t dt$$

► **Attempt a solution.**

Solving, we find that

$$a_0 = \frac{V_m}{\pi} \quad [14]$$

$$a_n = 5V_m \int_{-0.1}^{0.1} \frac{1}{2} [\cos 5\pi(1+n)t + \cos 5\pi(1-n)t] dt$$

or

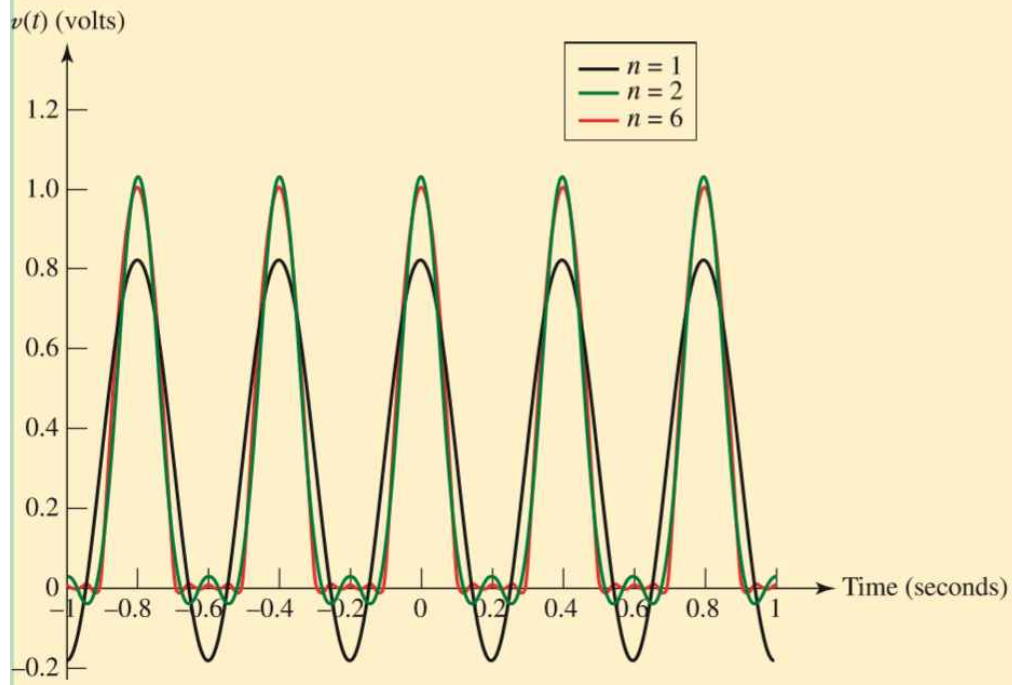
$$a_n = \frac{2V_m}{\pi} \frac{\cos(\pi n/2)}{1-n^2} \quad (n \neq 1) \quad [15]$$

(A similar integration shows that $b_n = 0$ for any value of n , and the Fourier series thus contains no sine terms.) The Fourier series is therefore obtained from Eqs. [1], [13], [14], and [15]:

$$\begin{aligned} v(t) = & \frac{V_m}{\pi} + \frac{V_m}{2} \cos 5\pi t + \frac{2V_m}{3\pi} \cos 10\pi t - \frac{2V_m}{15\pi} \cos 20\pi t \\ & + \frac{2V_m}{35\pi} \cos 30\pi t - \dots \end{aligned} \quad [16]$$

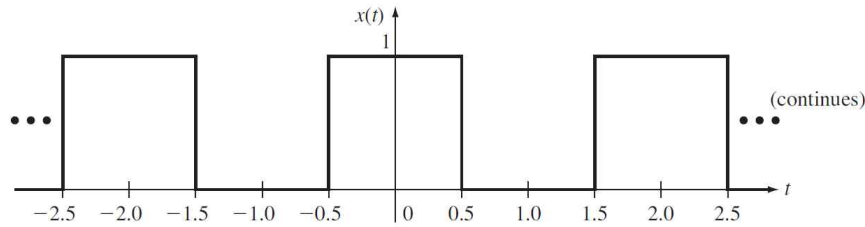
► **Verify the solution. Is it reasonable or expected?**

Our solution can be checked by plugging values into Eq. [16] and truncating after a specific number of terms. Another approach, however, is to plot the function as shown in Fig. 18.3 for $n = 1, 2$, and 6.



※ Circuit and Lab Textbook (END)

(※ Notations are different.)

(Example) Rectangular Pulse Train [* Kamen book *]

Consider the rectangular pulse train shown in Figure 3.5. This signal is periodic with fundamental period $T = 2$, and thus the fundamental frequency is $\omega_0 = 2\pi/2 = \pi$ rad/sec. The signal obviously satisfies the Dirichlet conditions, and thus it has a Fourier series representation. From (3.7), the constant component of $x(t)$ is

$$a_0 = \frac{1}{2} \int_0^2 x(t) dt = \frac{1}{2} \int_0^{0.5} (1) dt + \frac{1}{2} \int_{1.5}^2 (1) dt = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Evaluating (3.5) gives

$$\begin{aligned} a_k &= \frac{2}{2} \int_0^2 x(t) \cos(k\pi t) dt \\ &= \int_0^{0.5} \cos(k\pi t) dt + \int_{1.5}^2 \cos(k\pi t) dt \\ &= \frac{1}{\pi k} \sin(k\pi t) \Big|_{t=0}^{t=0.5} + \frac{1}{\pi k} \sin(k\pi t) \Big|_{t=1.5}^{t=2} \\ &= \frac{1}{\pi k} \left[\sin\left(\frac{k\pi}{2}\right) - \sin\left(\frac{3k\pi}{2}\right) \right] \\ &= \frac{1}{\pi k} \left[\sin\left(\frac{k\pi}{2}\right) - \sin\left(\frac{k\pi}{2} + k\pi\right) \right] \\ &= \frac{1}{\pi k} \left[2 \sin\left(\frac{k\pi}{2}\right) \right], k = 1, 2, \dots \end{aligned}$$

Evaluating (3.6) gives

$$\begin{aligned} b_k &= \frac{2}{2} \int_0^2 x(t) \sin(k\pi t) dt \\ &= \int_0^{0.5} \sin(k\pi t) dt + \int_{1.5}^2 \sin(k\pi t) dt \\ &= -\frac{1}{\pi k} \cos(k\pi t) \Big|_{t=0}^{t=0.5} - \frac{1}{\pi k} \cos(k\pi t) \Big|_{t=1.5}^{t=2} \\ &= -\frac{1}{\pi k} \left[\frac{\pi}{2} - \frac{\pi}{2} \right] \\ &= 0, k = 1, 2, \dots \end{aligned}$$

Then inserting the values for a_0 , a_k , and b_k into (3.4) results in the following Fourier series representation of the pulse train shown in Figure 3.5:

$$x(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{k} \sin\left(\frac{k\pi}{2}\right) \cos(k\pi t), \quad -\infty < t < \infty \quad (3.11)$$

Note that, since $\sin\left(\frac{k\pi}{2}\right) = 0$ for $k = 2, 4, 6, \dots$, the signal $x(t)$ contains only odd harmonics, and thus (3.11) can be written in the form

$$x(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{\substack{k=1 \\ k \text{ odd}}}^{\infty} \frac{1}{k} \sin\left(\frac{k\pi}{2}\right) \cos(k\pi t), \quad -\infty < t < \infty \quad (3.12)$$

Therefore,

$$x(t) = \frac{1}{2} + \frac{2}{\pi} \cos(\pi t) - \frac{2}{3\pi} \cos(3\pi t) + \frac{2}{5\pi} \cos(5\pi t) - \frac{2}{7\pi} \cos(7\pi t) + \dots$$

(* Euler's Identities)

$$\sin \alpha = \frac{e^{j\alpha} - e^{-j\alpha}}{j2}$$

$$\cos \alpha = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$$

$$e^{\pm j\alpha} = \cos \alpha \pm j \sin \alpha$$

§3.3.1 Linear combinations of harmonically related complex exponentials

- A periodic signal with period $T=2\pi/\omega_0$ can be represented by using the linear combination of the basic signals, $\cos m\omega_0 t$ ($m=0,1,2,\dots$) and/or $\sin n\omega_0 t$ ($n=0,1,2,\dots$). (Eq.(Δ))

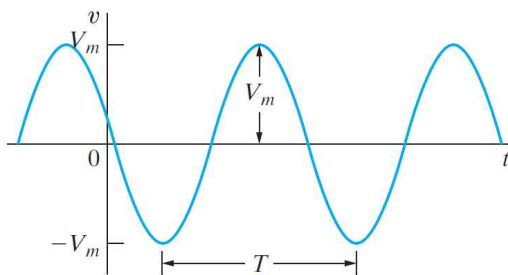
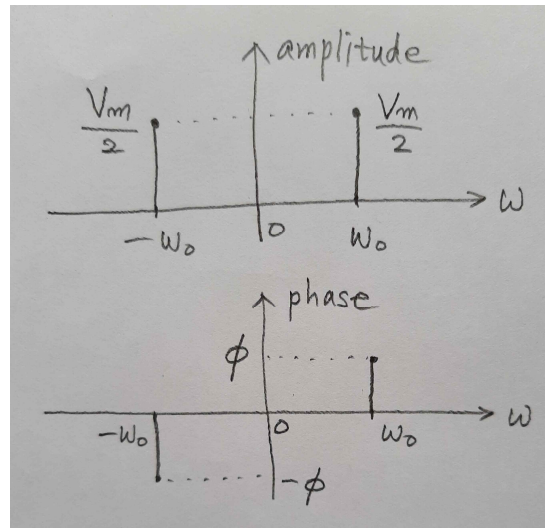
● Trigonometric Form to Exponential Form

$$\begin{aligned}
 x(t) &= a_0 + a_1 \cos \omega t + a_2 \cos 2\omega t + a_3 \cos 3\omega t + \dots \\
 &\quad + b_1 \sin \omega t + b_2 \sin 2\omega t + b_3 \sin 3\omega t + \dots \\
 &= a_0 + a_1 \frac{e^{j\omega t} + e^{-j\omega t}}{2} + a_2 \frac{e^{j2\omega t} + e^{-j2\omega t}}{2} + a_3 \frac{e^{j3\omega t} + e^{-j3\omega t}}{2} + \dots \\
 &\quad + b_1 \frac{e^{j\omega t} - e^{-j\omega t}}{2j} + b_2 \frac{e^{j2\omega t} - e^{-j2\omega t}}{2j} + b_3 \frac{e^{j3\omega t} - e^{-j3\omega t}}{2j} + \dots \\
 &= a_0 + \frac{(a_1 - jb_1)}{2} e^{j\omega t} + \frac{a_2 - jb_2}{2} e^{j2\omega t} + \frac{a_3 - jb_3}{2} e^{j3\omega t} + \dots \\
 &\quad + \frac{a_1 + jb_1}{2} e^{-j\omega t} + \frac{a_2 + jb_2}{2} e^{-j2\omega t} + \frac{a_3 + jb_3}{2} e^{-j3\omega t} + \dots \\
 &= a_0 + \sum_{n=1}^{\infty} \frac{a_n - jb_n}{2} e^{jn\omega t} + \sum_{n=1}^{\infty} \frac{a_n + jb_n}{2} e^{-jn\omega t} \\
 \text{Eq[10]} \quad a_n &= \frac{2}{T} \int_0^T x(t) \cos n\omega t dt \Rightarrow a_{-n} = a_n \\
 \text{Eq[11]} \quad b_n &= \frac{2}{T} \int_0^T x(t) \sin n\omega t dt \Rightarrow b_{-n} = -b_n \\
 &= a_0 + \sum_{n=1}^{\infty} \frac{a_n - jb_n}{2} e^{jn\omega t} + \sum_{m=-1}^{-\infty} \frac{a_{-m} + jb_{-m}}{2} e^{jm\omega t} \\
 &= a_0 + \sum_{n=1}^{\infty} \frac{a_n - jb_n}{2} e^{jn\omega t} + \sum_{n=-1}^{-\infty} \frac{a_{-n} + jb_{-n}}{2} e^{jn\omega t} \\
 &= a_0 + \sum_{n=1}^{\infty} \frac{a_n - jb_n}{2} e^{jn\omega t} + \sum_{n=-1}^{-\infty} \frac{a_n - jb_n}{2} e^{jn\omega t} \\
 &= a_0 + \sum_{\substack{n=-\infty \\ (n \neq 0)}}^{\infty} \frac{a_n - jb_n}{2} e^{jn\omega t} \\
 &= \sum_{n=-\infty}^{\infty} C_n e^{jn\omega t} \quad \left\{ \begin{array}{l} C_n = \frac{a_n - jb_n}{2}, C_0 = a_0 \end{array} \right.
 \end{aligned}$$

A Representation Method of Sinusoidal Wave

$$v(t) = V_m \cos(\omega_0 t + \phi) = \frac{V_m}{2} [e^{j(\omega_0 t + \phi)} + e^{-j(\omega_0 t + \phi)}]$$

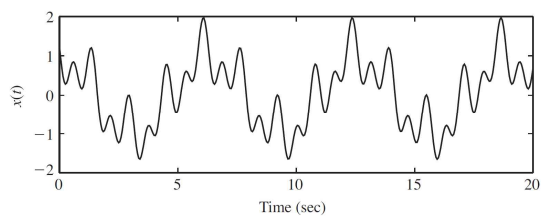
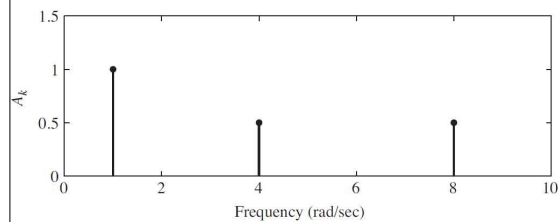
time domain

frequency domain
(spectrum)

(example) a summation of sinusoidal waves

$$x(t) = 1 \cos(t) + 0.5 \cos(4t + \pi/3) + 0.5 \cos(8t + \pi/2)$$

time domain

frequency domain
(Amplitude Spectra)

Trigonometric Form of Fourier Series (Not widely used)

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \quad [1]$$

$$\omega_0 = \frac{2\pi}{T} = 2\pi f_0$$

$$a_0 = \frac{1}{T} \int_0^T f(t) dt \quad [9]$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos n\omega_0 t dt \quad [10]$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega_0 t dt \quad [11]$$

!!! Notations !!!

(other textbook)

trigonometric form	$ \begin{aligned} x(t) &= a_0 + a_1 \cos \omega_0 t + a_2 \cos 2\omega_0 t + \dots \\ &\quad + b_1 \sin \omega_0 t + b_2 \sin 2\omega_0 t + \dots \\ &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\ &= a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t - \theta_n) \end{aligned} $
exponential form	$x(t) = \sum_{n=-\infty}^{+\infty} c_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{+\infty} c_n e^{jn(2\pi/T)t}$

(our textbook)

trigonometric form	$ \begin{aligned} x(t) &= a_0 + 2B_1 \cos \omega_0 t + 2B_2 \cos 2\omega_0 t + \dots \\ &\quad - 2C_1 \sin \omega_0 t - 2C_2 \sin 2\omega_0 t + \dots \\ &= a_0 + \sum_{k=1}^{\infty} (2B_k \cos k\omega_0 t - 2C_k \sin k\omega_0 t) \\ &= a_0 + \sum_{k=1}^{\infty} 2A_k \cos(k\omega_0 t + \theta_k) \end{aligned} $
exponential form	$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$

- For real periodic functions, the Fourier series in terms of complex exponentials is mathematically equivalent to the Fourier series in terms of trigonometric functions.
- The complex exponential form of the Fourier series is particularly convenient for our purpose, so we will use the complex exponential form of the Fourier series from now.

- CONTINUOUS TIME FOURIER SERIES (CTFS)

A linear combination of harmonically related complex exponentials of the form is also periodic with period T .

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t} \quad (3.25)$$

$e^{j\omega_0 t}, e^{-j\omega_0 t}$: the fundamental components
or the first harmonic components
 $e^{j2\omega_0 t}, e^{-j2\omega_0 t}$: the second harmonic components
 $e^{jN\omega_0 t}, e^{-jN\omega_0 t}$: the Nth harmonic components

(Example 3.2)

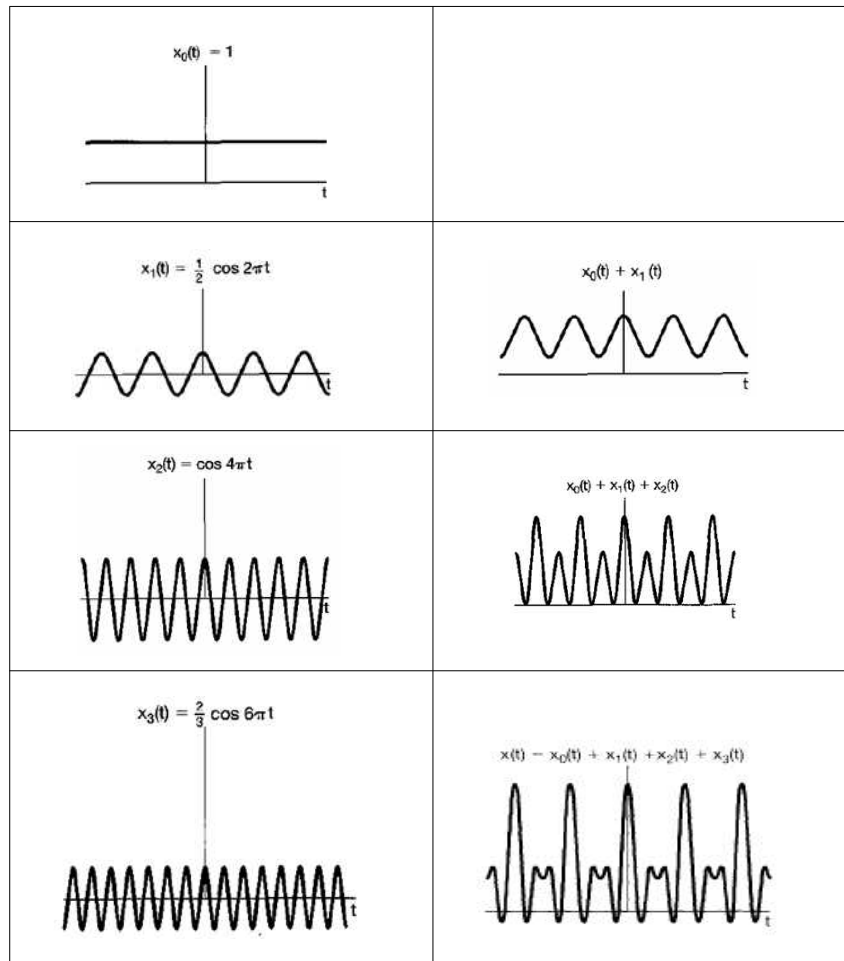
Consider a periodic signal $x(t)$ with $\omega_0 = 2\pi$ [rad/sec]

$$x(t) = \sum_{k=-3}^{+3} a_k e^{jk2\pi t} \quad (3.26)$$

where $a_0 = 1$, $a_1 = a_{-1} = 1/4$, $a_2 = a_{-2} = 1/2$, $a_3 = a_{-3} = 1/3$.

Rewriting eq.(3.26),

$$\begin{aligned} x(t) &= 1 + \frac{1}{4}(e^{j2\pi t} + e^{-j2\pi t}) + \frac{1}{2}(e^{j4\pi t} + e^{-j4\pi t}) + \frac{1}{3}(e^{j6\pi t} + e^{-j6\pi t}) \\ &= 1 + \frac{1}{2} \cos 2\pi t + \cos 4\pi t + \frac{2}{3} \cos 6\pi t \end{aligned}$$



Specifically, suppose that $x(t)$ is real, then $x^*(t) = x(t)$. Thus,

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k^* e^{-jk\omega_0 t}$$

Replacing k by $-k$ in this summation,

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k^* e^{-jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_{-k}^* e^{jk\omega_0 t}$$

Therefore,

$$a_k^* = a_{-k}$$

- Alternative form of the Fourier series

$$\begin{aligned}
 x(t) &= \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \\
 &= a_0 + \sum_{k=1}^{\infty} [a_k e^{jk\omega_0 t} + a_{-k} e^{-jk\omega_0 t}] \\
 &= a_0 + \sum_{k=1}^{\infty} [a_k e^{jk\omega_0 t} + a_k^* e^{-jk\omega_0 t}] \\
 &= a_0 + \sum_{k=1}^{\infty} 2 \operatorname{Re}\{a_k e^{jk\omega_0 t}\} \tag{3.30}
 \end{aligned}$$

If a_k is expressed in polar form as

$$a_k = A_k e^{j\theta_k} ,$$

then Eq.(3.30) becomes

$$\begin{aligned}
 x(t) &= a_0 + \sum_{k=1}^{\infty} 2 \operatorname{Re}\{A_k e^{j(k\omega_0 t + \theta_k)}\} \\
 &= a_0 + 2 \sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k) \tag{3.31}
 \end{aligned}$$

- Another form of the Fourier series

If a_k is expressed in rectangular form as

$$a_k = B_k + j C_k ,$$

then Eq.(3.30) becomes

$$x(t) = a_0 + 2 \sum_{k=1}^{\infty} [B_k \cos k\omega_0 t - C_k \sin k\omega_0 t] \tag{3.32}$$

For real periodic functions, the Fourier series in terms of complex exponentials [Eq.(3.25)] is mathematically equivalent to either of the two forms in Eqs.(3.31) and (3.32) that use trigonometric functions.

Although the two forms in Eqs.(3.31) and (3.32) are common forms for Fourier series [Circuit Theory], the complex exponential form of Eq.(3.25) is particularly convenient for our purpose, so we will use that form almost exclusively.

§3.3.2 Determination of the Fourier coefficients

- Assuming that a given periodic signal $x(t)$ with the fundamental frequency ω_0 can be represented with the series of eq (3.25), we need a procedure for determining the coefficients a_k [Fourier coefficients]

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t} \quad (3.25)$$

Multiplying both sides of Eq.(3.25) by $e^{-jn\omega_0 t}$,

$$x(t) e^{-jn\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} e^{-jn\omega_0 t} \quad (3.33)$$

and integrating both sides from 0 to T

$$\begin{aligned} \int_0^T x(t) e^{-jn\omega_0 t} dt &= \int_0^T \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} e^{-jn\omega_0 t} dt \\ &= \sum_{k=-\infty}^{\infty} a_k \left[\int_0^T e^{j(k-n)\omega_0 t} dt \right] \end{aligned} \quad (3.34)$$

Using Euler's formula,

$$\int_0^T e^{j(k-n)\omega_0 t} dt = \int_0^T \cos(k-n)\omega_0 t dt + j \int_0^T \sin(k-n)\omega_0 t dt \quad (3.35)$$

So,

$$\int_0^T e^{j(k-n)\omega_0 t} dt = \begin{cases} T, & k=n \\ 0, & k \neq n \end{cases}$$

Therefore, we obtain from Eq.(3.34),

$$a_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt \quad (3.36)$$

– Continuous-Time Fourier Series (CTFS) Pair

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t} \quad (3.38)$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_T x(t) e^{-jk(2\pi/T)t} dt \quad (3.39)$$

Eq.(3.38) : synthesis equation

Eq.(3.39) : analysis equation

$\{a_k\}$: Fourier series coefficients or spectral coefficients of $x(t)$

a_0 : DC component of $x(t)$ ($a_0 = \frac{1}{T} \int_0^T x(t) dt$) [average value]

※ Eqs.(3.38) and (3.39) were known to both Euler and Lagrange in the middle of the 18th century. However, they discarded this line of analysis without having examined the question of how large a class of periodic signals could be represented in such a fashion.

(Example 3.4)

$$\begin{aligned}
 x(t) &= 1 + \sin \omega_0 t + 2 \cos \omega_0 t + \cos(2 \omega_0 t + \pi/4) \\
 &= 1 + \left(1 + \frac{1}{2j}\right) e^{j\omega_0 t} + \left(1 - \frac{1}{2j}\right) e^{-j\omega_0 t} + \left(\frac{1}{2} e^{j(\pi/4)}\right) e^{j2\omega_0 t} + \left(\frac{1}{2} e^{-j(\pi/4)}\right) e^{-j2\omega_0 t} \\
 &\equiv \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}
 \end{aligned}$$

Thus, the Fourier series coefficients

$$a_0 = 1$$

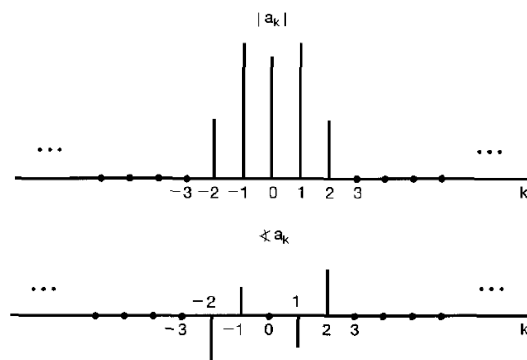
$$a_1 = 1 + \frac{1}{2j} = 1 - \frac{1}{2} j$$

$$a_{-1} = 1 - \frac{1}{2j} = 1 + \frac{1}{2} j$$

$$a_2 = \frac{1}{2} e^{j(\pi/4)} = \frac{\sqrt{2}}{4} (1 + j)$$

$$a_{-2} = \frac{1}{2} e^{-j(\pi/4)} = \frac{\sqrt{2}}{4} (1 - j)$$

$$a_k = 0, \quad |k| > 2$$

a bar graph of the magnitude and phase of a_k [the spectrum of $x(t)$]

※ other representation method for a sinewave using 2 bars,
that is, the representation method in the frequency domain.

(Example 3.5) a periodic square wave

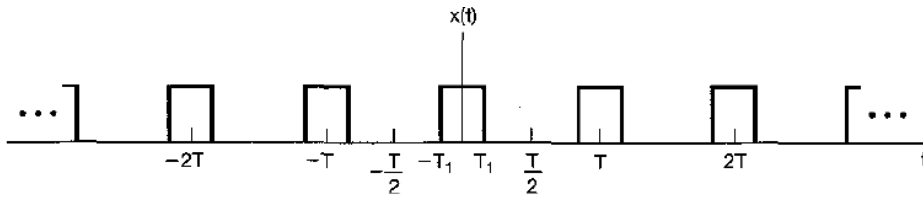


Figure 3.6 Periodic square wave.

For $k = 0$,

$$a_0 = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) dt = \frac{1}{T} \int_{-T_1}^{T_1} 1 dt = \frac{2T_1}{T}$$

For $k \neq 0$,

$$\begin{aligned} a_k &= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-T_1}^{T_1} 1 e^{-jk\omega_0 t} dt \\ &= -\frac{1}{jk\omega_0 T} e^{-jk\omega_0 t} \Big|_{-T_1}^{T_1} \\ &= \frac{2}{k\omega_0 T} \left[\frac{e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}}{2j} \right] \\ &= \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{\sin(k\omega_0 T_1)}{k\pi}, \quad k \neq 0 \end{aligned} \quad (3.44)$$

Plots of the Fourier series coefficients Ta_k

$T = 4T_1$ (period)=2(pulse width)	
$T = 8T_1$ (period)=4(pulse width)	
$T = 16T_1$ (period)=8(pulse width)	

the envelope : $(2 \sin \omega T_1) / \omega$

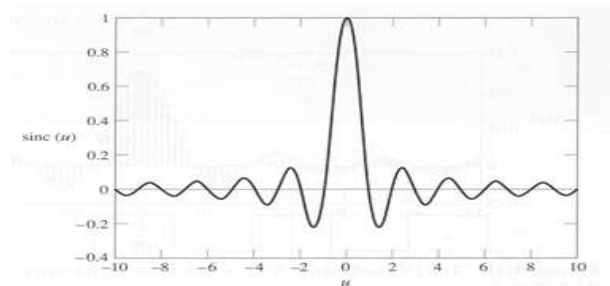
the spacing between samples : $2\pi / T$

the first zero-crossing : 2nd / 4th / 8th

※ Since $a_0 = \lim_{k \rightarrow 0} \frac{2 \sin(k \omega_0 T_1)}{k \omega_0 T} = \frac{2 T_1}{T}$,

a_k can be represented $a_k = \frac{2 \sin(k \omega_0 T_1)}{k \omega_0 T}$

※ sinc function : $\text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}$



main lobe : between the first zero crossings

side lobes : outsides of the first zero crossing

※ Eq.(3.44) can be represented using sinc function

$$a_k = \frac{2 \sin(2k\pi \frac{T_1}{T})}{2k\pi} = \frac{2 T_1}{T} \frac{\sin(2k\pi \frac{T_1}{T})}{2\pi k \frac{T_1}{T}} = \frac{2 T_1}{T} \text{sinc}(2k \frac{T_1}{T})$$

$$= \frac{\tau}{T} \frac{\sin(k\pi \frac{\tau}{T})}{k\pi \frac{\tau}{T}} = \frac{\tau}{T} \text{sinc}(k \frac{\tau}{T})$$

(let $\tau = 2 T_1$)

τ : pulse width

(※ Duty Cycle $\frac{\tau}{T} \times 100$ [%])